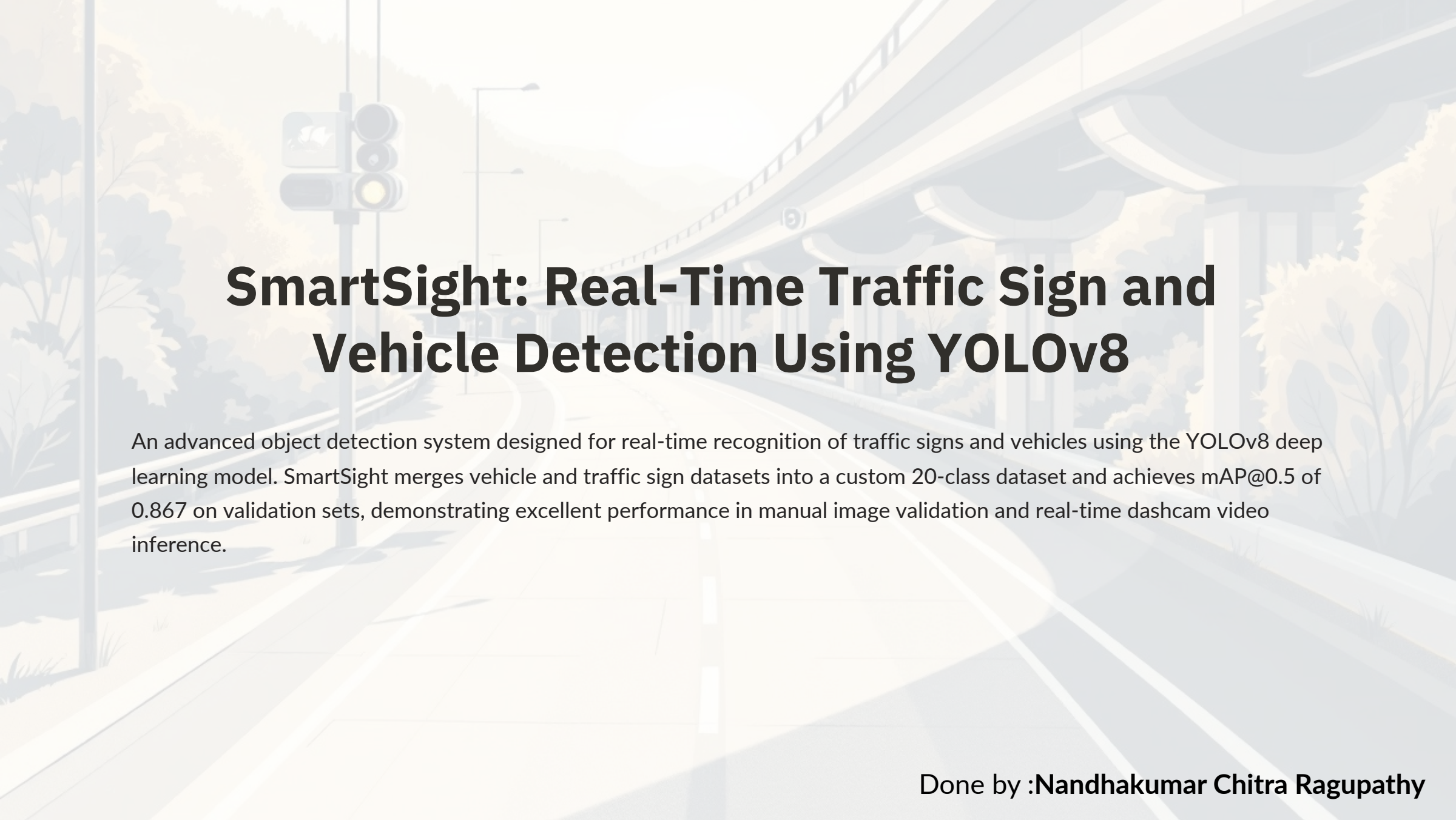


The background is a stylized, light-colored illustration of a road scene. On the left, there is a traffic light pole with a traffic light showing a yellow light. The road curves to the right, leading towards a bridge with multiple arches in the distance. There are some trees and bushes on the left side of the road. The overall style is clean and modern, with a focus on the road and infrastructure.

SmartSight: Real-Time Traffic Sign and Vehicle Detection Using YOLOv8

An advanced object detection system designed for real-time recognition of traffic signs and vehicles using the YOLOv8 deep learning model. SmartSight merges vehicle and traffic sign datasets into a custom 20-class dataset and achieves mAP@0.5 of 0.867 on validation sets, demonstrating excellent performance in manual image validation and real-time dashcam video inference.

Done by :Nandhakumar Chitra Ragupathy

Introduction: The Critical Role of Automated Detection

The ability to detect vehicles and traffic signs automatically is fundamental for autonomous vehicles, driver assistance systems, and traffic safety monitoring infrastructure. Accurate object detection improves navigation capabilities, reduces accident rates, and supports intelligent traffic management systems in modern urban environments.

Traditional computer vision methods were severely limited in processing speed and detection accuracy, particularly in complex real-world scenarios. Deep learning-based models, especially YOLOv8, provide high-speed, anchor-free, multi-class detection capabilities that are ideally suited for dynamic traffic scenarios requiring real-time processing.

867
mAP@0.5

Validation accuracy

28

FPS

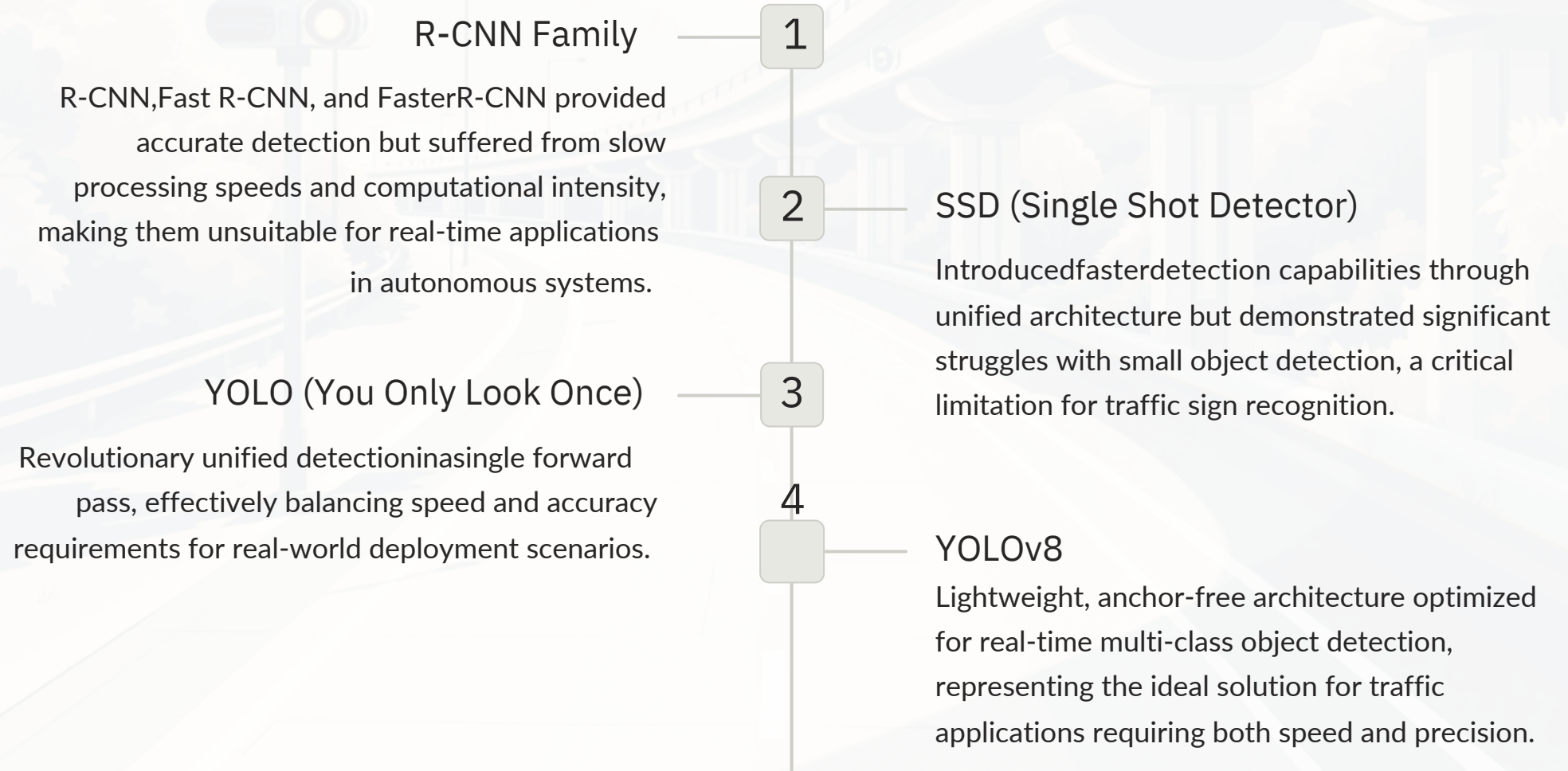
Real-time processing

20

Classes

Detection categories

Literature Review: Evolution of Object Detection



Dataset Architecture and Structure

Dataset Organization

The SmartSight system utilizes a comprehensive dataset structure combining vehicle detection and traffic sign recognition into a unified 20-class dataset. Images are annotated in YOLO format with normalized coordinates (class, x_center, y_center, width, height) ensuring consistent preprocessing across all training phases.

Split	Purpose
Train	Model weight optimization Hyperparameter tuning
Validation	Final performance evaluation
Test	

Data Augmentation Pipeline

Advanced augmentation techniques enhance model robustness and generalization capabilities across diverse traffic scenarios and environmental conditions.



Geometric Transforms

Random rotations, scaling, and resizing operations



Mosaic Augmentation

Multi-image composition for complex scene learning



Photometric Adjustments

Brightness and contrast variation simulation



Detection Classes: Comprehensive Coverage

Vehicle Categories

- Ambulance
- Bus
- Car
- Motorcycle
- Truck

Traffic Signals

- Green Light
- Red Light
- Stop Sign

Speed Limits

- 10, 20, 30, 40, 50 km/h
- 60, 70, 80, 90 km/h
- 100, 110, 120 km/h

The 20-class detection system provides comprehensive coverage of critical traffic elements, enabling robust performance across diverse driving scenarios and regulatory environments. This classification scheme balances granularity with computational efficiency.

Methodology: YOLOv8 Implementation

ModelArchitecture

SmartSight leverages thepre-trained YOLOv8.pt base model within the Ultralytics PyTorch framework. The architecture employs anchor-free detection with sophisticated loss functions including box regression, classification, and Distribution Focal Loss (DFL) for enhanced small object detection.



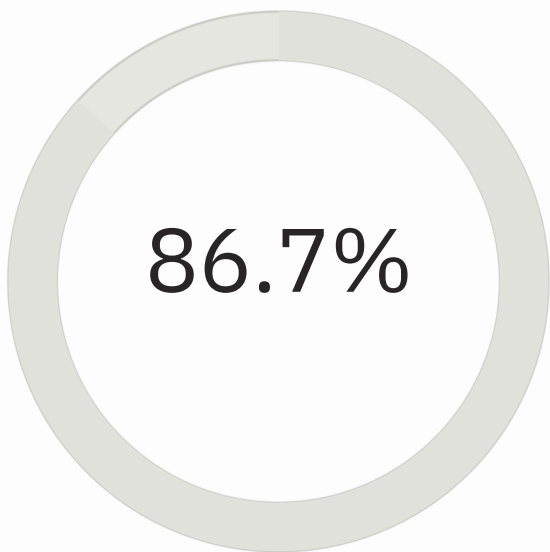
The anchor-free approach eliminates manual anchor design, enabling more flexible detection across varying object scales and aspect ratios common in traffic environments.

Training Configuration

Parameter	Value
Epochs	100
Batch Size	8
Learning Rate	0.001
Optimizer	AdamW
Scheduler	Cosine decay
Image Size	640×640 px
Hardware	RTX 4050 (6GB)

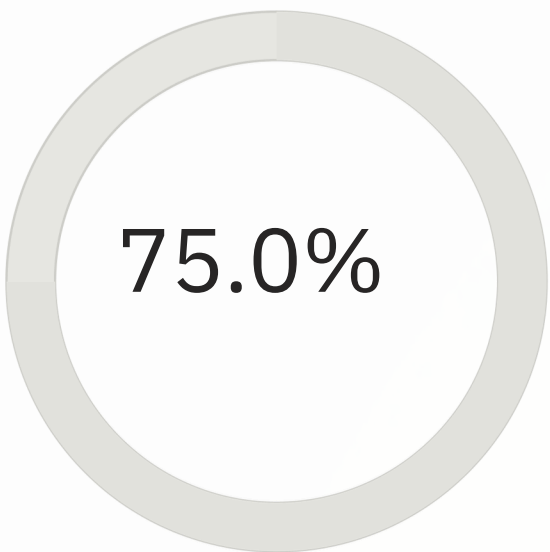
📄 **Saved Weights:** best.pt (optimal validation) and last.pt (final epoch) enable flexible deployment strategies.

Evaluation Metrics and Validation Results



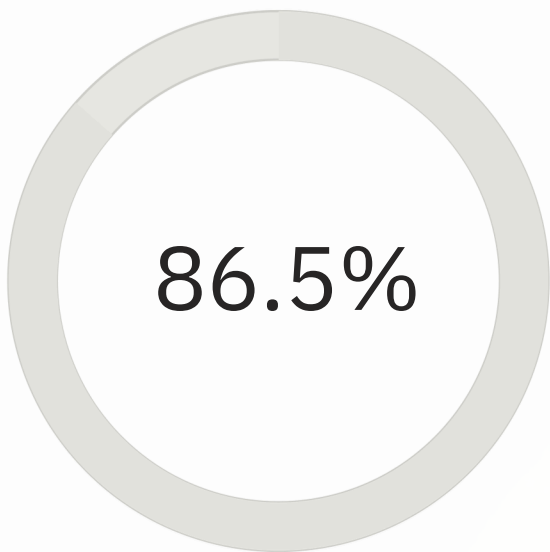
mAP@0.5

Primary accuracy metric



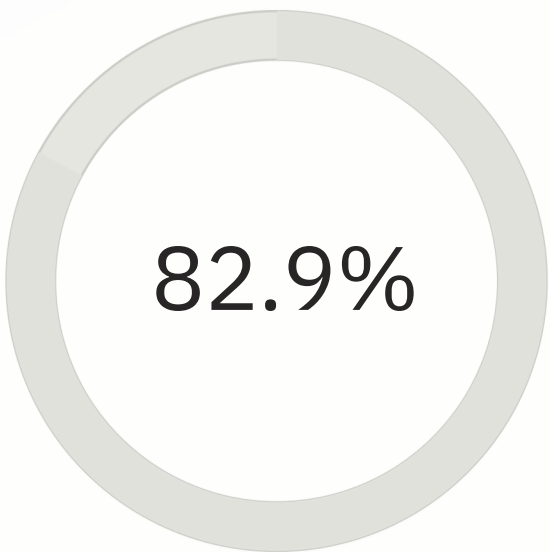
mAP@0.5:0.95

Strict IoU threshold



Mean Precision

Detection confidence



Mean Recall

Object coverage rate

Class-Wise Performance Analysis

Exceptional Performance

Stopsigns and speedlimit signs achieved mAP exceeding 0.99, demonstrating near-perfect detection accuracy for regulatory traffic signage.

Traffic Light Detection

Red lights: 0.807 mAP
Green lights: 0.855 mAP
Strong signal detection across lighting conditions.

Vehicle Categories

Cars and trucks showed moderate performance (~0.42-0.39 mAP), indicating opportunities for targeted improvement.

Manual Validation and Real-Time Inference

OpenCV Manual Validation Pipeline

Manual inspection using OpenCV verified predictions through visual bounding box and label rendering. The workflow loaded test images, executed YOLOv8 predictions, drew detection boxes using `cv2.rectangle` and `cv2.putText`, then resized images for comprehensive visual verification.

1

Ambulance Detection

Single object detection with 28.6 ms inference time, confirming emergency vehicle recognition capabilities.

2

Multi-Bus Scenario

Three bus detections with 28.8-34.7 ms processing, validating crowded scene performance.

3

Motorcycle Recognition

32-33.1 ms inference demonstrating small vehicle detection accuracy under motion.

4

Stop Sign Precision

19.0 ms inference achieving optimal regulatory sign detection speed and accuracy.

Real-Time Video Performance Analysis

Dashcam Processing

Real-time inference on dashcam video (384×640 resolution) processed 3,934 frames with 0.25 confidence threshold. The system detected multiple vehicles and traffic lights per frame while maintaining 28 FPS average processing speed.

01

Preprocessing: 7.1 ms

Image normalization and formatting

02

Inference: 35.3 ms

Neural network forward pass

03

Post-processing: 5.6 ms

NMS and result formatting



Key Observations

The system demonstrated robust performance under motion blur and varying lighting conditions, confirming deployment readiness for real-world traffic monitoring applications. Output videos saved to runs/detect/predict enable comprehensive quality assessment.

YOLOv8's lightweight architecture ensures deployment potential on edge devices while maintaining detection accuracy across challenging environmental conditions and complex traffic scenarios.



Conclusions and Future Development

Key Achievements

SmartSight successfully detects traffic signs and vehicles in images and video with high accuracy and real-time performance. The system provides a scalable, robust solution suitable for intelligent traffic management, autonomous driving assistance, and comprehensive safety monitoring applications across diverse deployment scenarios.



Multi-Object Tracking

Integrate DeepSORT or ByteTrack for persistent vehicle tracking across frames



Weather Robustness

Expand training data to include adverse weather conditions



Edge Deployment

Optimize for Jetson Nano and embedded systems



Enhanced Features

Add license plate recognition and vehicle counting capabilities